

高中基础知识小测

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1 数学

1.1 集合与逻辑

集合 A 中有 n 个元素, 则 A 有_____个子集, _____个真子集/非空子集, _____个非空真子集

P 是 Q 的子集, 则 P 是 Q 的_____条件

1.2 函数与方程

基本不等式:

$$\underline{\hspace{2cm}} \leq \underline{\hspace{2cm}} \leq \underline{\hspace{2cm}} \leq \underline{\hspace{2cm}}$$

$$\log_a b \times \underline{\hspace{2cm}} = 1$$

画图:

$$f(x) = a^x$$

$$g(x) = x^a$$

$$h(x) = \log_a x$$

反函数是什么:

极值与最值都是什么:

三角恒等变换:

$$\sin\left(\frac{\alpha}{2}\right) = \underline{\hspace{2cm}} \quad \cos\left(\frac{\alpha}{2}\right) = \underline{\hspace{2cm}} \quad \tan\left(\frac{\alpha}{2}\right) = \underline{\hspace{2cm}} \text{ (带根号)} = \underline{\hspace{2cm}} \text{ (不带)}$$

$$\text{用 tan 表示:} \quad \sin(2\alpha) = \underline{\hspace{2cm}} \quad \cos(2\alpha) = \underline{\hspace{2cm}}$$

$$y = a \sin(x) + b \cos(x) = \underline{\hspace{2cm}} \quad y = a \sin(x) - b \cos(x) = \underline{\hspace{2cm}}$$

$$y = -a \sin(x) + b \cos(x) = \underline{\hspace{2cm}} \quad y = -a \sin(x) - b \cos(x) = \underline{\hspace{2cm}}$$

$$\sin(\alpha) \cos(\beta) = \underline{\hspace{2cm}} \quad \cos(\alpha) \cos(\beta) = \underline{\hspace{2cm}}$$

$$\cos(\alpha) \sin(\beta) = \underline{\hspace{2cm}} \quad \sin(\alpha) \sin(\beta) = \underline{\hspace{2cm}}$$

$$\sin(\alpha) + \sin(\beta) = \underline{\hspace{2cm}} \quad \cos(\alpha) + \cos(\beta) = \underline{\hspace{2cm}}$$

$$\sin(\alpha) - \sin(\beta) = \underline{\hspace{2cm}}$$

$$\cos(\alpha) - \cos(\beta) = \underline{\hspace{2cm}}$$